

Chi-Square Tests for the Final Exam

MAT 1272 Statistics

Learning Objectives

By the end of this review, you should be able to:

- Identify a Chi-Square Goodness-of-Fit Test.
- Identify a Chi-Square Test of Independence.
- State the null and alternative hypotheses.
- Compute the degrees of freedom.
- Recognize that chi-square tests are right-tailed.
- Use the TI-84 to obtain a p-value.
- Write a conclusion in context.

Two Types of Chi-Square Tests

1. Goodness-of-Fit Test

- One categorical variable.
- Compare observed counts to expected percentages.

2. Test of Independence

- Two categorical variables.
- Data are presented in a contingency table.

Goodness-of-Fit Test

Use this test when there is only one categorical variable.

Examples:

- Cereal brands
- Causes of forest fires
- Favorite sports
- Political affiliation

Hypotheses:

H_0 : The distribution is as claimed.

H_1 : The distribution is different.

Degrees of Freedom for Goodness-of-Fit

Formula:

$$df = n - 1$$

where n is the number of categories.

Example:

Four cereal brands

$$df = 4 - 1 = 3$$

Test of Independence

Use this test when there are two categorical variables.

Example:

	Yes	No	No Opinion
Borough B	463	86	31
Borough M	295	189	36

Hypotheses:

H_0 : The variables are independent.

H_1 : The variables are dependent.

Degrees of Freedom for Independence

Formula:

$$df = (R - 1)(C - 1)$$

where

- R = number of rows
- C = number of columns

Example:

$$df = (2 - 1)(3 - 1) = 2$$

Chi-Square Tests Are Always Right-Tailed

Students should memorize:

Every Chi-Square Test is Right-Tailed.

The rejection region is always located in the right tail.

Good to keep in mind for the Exam

For every Chi-Square question:

- 1 State H_0 and H_1 .
- 2 Identify the claim.
- 3 Find the degrees of freedom.
- 4 State that the test is right-tailed.
- 5 Find the p-value.
- 6 Make a decision.
- 7 Write a conclusion.

Goodness-of-Fit

- 1 Enter observed counts in L_1 .
- 2 Enter expected counts in L_2 .
- 3 STAT
- 4 TESTS
- 5 χ^2 -GOF Test
- 6 Read the p-value.

Test of Independence

- 1 Enter the contingency table in Matrix A.
- 2 STAT
- 3 TESTS
- 4 χ^2 -Test
- 5 Read the p-value.

Recipe-Style Goodness-of-Fit Example

Healthy Food Company cereal:

Brand	A	B	C	D
Count	212	284	254	250

$$\alpha = 0.05$$

Step 1

H_0 : The distribution is as claimed

H_1 : The distribution is different

Step 2

$$df = 4 - 1 = 3$$

Step 3

Use the TI-84 χ^2 -GOF Test.

Step 4

Read the p-value.

Step 5

Compare the p-value with α .

Step 6

Write a conclusion in context.

Review Worksheet Question 26

A survey produced the following results.

	Yes	No	No Opinion
Borough B	463	86	31
Borough M	295	189	36

Use

$$\alpha = 0.01$$

to test whether borough and opinion are independent.

Step 1: Hypotheses

H_0 : Borough and opinion are independent

H_1 : Borough and opinion are dependent

Step 2: Identify the claim

The claim is H_0 .

Step 3: Degrees of freedom

$$df = (2 - 1)(3 - 1) = 2$$

Step 4: Tail

All chi-square tests are right-tailed.

Step 5: Calculator

Use the TI-84 χ^2 -Test.

Suppose the calculator gives

$$p = 1.32 \times 10^{-16}$$

Step 6: Decision

$$p < \alpha$$

$$1.32 \times 10^{-16} < 0.01$$

Reject H_0 .

Step 7: Conclusion

There is sufficient evidence that borough and opinion are dependent.

Summary

For the Final Exam:

- Identify the Chi-Square test.
- State H_0 and H_1 .
- Find the degrees of freedom.
- Chi-Square tests are always right-tailed.
- Use the TI-84 to obtain the p-value.
- Compare the p-value with α .
- Write a conclusion in context.

Good Luck!